E.G.S.PILLAY ARTS & SCIENCE COLLEGE
(AUTONOMOUS)
NAGAPATTINAM–611002. TAMILNADU
(Accredited by NAAC With "A" Grade)
(MBA, BBA & BCA–Approved by AICTE, New Delhi)
(Affiliated to Bharathidasan University,Tiruchirappalli–24)

B.Sc., DS – Bachelor of Science in Data Science

Syllabus

Choice Based Credit System-
Learning Outcomes Based Curriculum Framework
(CBCS-LOCF)

(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

B.Sc., DS-Bachelor of Science in Data Science
Choice Based Credit System Learning Outcomes Based Curriculum Framework (CBCS-LOCF)
(Applicable to the candidate admitted from the academic year 2025–2026 onwards)

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
I	Language Course – I (LC) –Tamil Other Languages	Tamil-I	6	3	3	25	75	100
	English Language Course -I(ELC)	English-I	6	3	3	25	75	100
	Core Course–I(CC)	Programming in C and Data Structures	6	5	3	25	75	100
	Core Practical-I(CP)	Programming in C Lab	4	4	3	50	50	100
	First Allied Course–I(AC)	Descriptive Statistics	6	4	3	25	75	100
	Value Education	Value Education	2	2	3	25	75	100
	Total		**30**	**21**				**600**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
II	Language Course – II (LC) -Tamil/Other Languages	Tamil-II	6	3	3	25	75	100
	English Language Course – II(ELC)	English-II	6	3	3	25	75	100
	Core Course–II(CC)	Programming in Python	5	4	3	25	75	100
	Core Practical-II(CP)	Programming in Python Lab	4	4	3	50	50	100
	First Allied Course-II (AC)	Inferential Statistics	4	3	3	25	75	100
	First Allied practical– I (AP)	Statistics for Data Science Lab	3	2	3	50	50	100
	Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total		**30**	**21**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
III	Language Course – III (LC) -Tamil/Other Languages	Tamil-III	6	3	3	25	75	100
	English Language Course -III (ELC)	English-III	6	3	3	25	75	100
	Core Course – III (CC)	Database Systems	6	5	3	25	75	100
	Core Practical - III (CP)	Database Systems Lab	4	4	3	50	50	100
	Second Allied Course – I (AC)	Mathematics-I	6	4	3	25	75	100
	Non Major Elective I	**A.** Fundamentals of Information Technology **B.** Basics of office automation	2	2	3	25	75	100
	Activity Based Course	Health & Wellness	-	1	-	-	-	100
	Total		**30**	**22**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
IV	Language Course –IV (LC) - Tamil/Other Languages	Tamil-IV	6	3	3	25	75	100
	English Language Course– IV (ELC)	English-IV	6	3	3	25	75	100
	Core Course–IV(CC)	Big Data Analytics with NoSQL Database	5	5	3	25	75	100
	Core Practical-IV(CP)	Big Data Analytics with NoSQL Database Lab	4	4	3	50	50	100
	Second Allied Course-II(AC)	Mathematics-II	4	3	3	25	75	100
	Second Allied Practical (AP)	Data Analytics Using Spread Sheets Lab	3	2	3	50	50	100
	Non Major Elective II	A. Working principles of Internet B. Web Designing	2	2	3	25	75	100
	Total		**30**	**22**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
V	Core Course V[CC]	R Programming and Data Visualization	6	5	3	25	75	100
	Core Course VI[CC]	Network Security	5	5	3	25	75	100
	Core Course VII[CC]	Cloud Computing	5	5	3	25	75	100
	Core Practical V[CP]	R Programming and Data Visualization Lab	5	4	3	50	50	100
	Major Based Elective-I	A. Internet of Things B. Block Chain Technology	5	4	3	25	75	100
	Skill Based Elective- I	A. Elements of Distributed Data Processing B. Digital Marketing	2	2	3	25	75	100
	Internship	Internship (Report of the Study- 80 Marks, Viva Voce- 20 Marks)	-	2	-	20	80	100
	Soft Skills Development	Soft Skills Development	2	2	3	25	75	100
	Total		**30**	**29**				**800**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
						Int.	Ext.	Total
VI	Core Course VIII[CC]	Data Mining and Warehousing	6	5	3	25	75	100
	Core Course IX [CC]	Machine Learning	6	5	3	25	75	100
	Core Practical VI[CP]	Machine Learning Lab	5	4	3	50	50	100
	Major Based Elective-II	A. Computing Technologies for Data Science B. Augmented Reality and Virtual Reality	5	4	3	25	75	100
	Project	Project (Project Dissertation- 80 Marks, Viva Voce- 20 Marks)	5	3	3	20	80	100
	Skill Based Elective -II	A. Social Media Analytics B. Business Intelligence	2	2	3	25	75	100
	Extension Activities	NCC,NSS,YRC,YWF	-	1	-	-	-	-
	Gender Studies	Gender Studies	1	1	3	25	75	100
	Total		**30**	**25**				**700**
	Grand Total		**180**	**140**				**4200**

18/8/25

R. Priya/18.8.25
DR. R. PRIYA
ASSISTANT PROFESSOR
SRMIST, CHENNAI

18/8/25
DR. K. UMAMAHESWARI
ASSISTANT PROFESSOR
AGICW, KARAIKAL

Dr. K. NANDHINI
Assistant Professor
Dept of Computer Science
Central University of Tamilnadu

18/8/25
BOS - CHAIRMAN
Department of Artificial Intelligence & Machine Learning, Data Science and Computer Science with Cognitive Systems,
E.G.S.Pillay Arts & Science College (Autonomous),
Nagapattinam - 611 002

B.Sc., DS-Bachelor of Science in Data Science
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	Core Course–I(CC)	Programming in C and Data Structures	6	5	3	25	75	100
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	Core Course – III (CC)	Database Systems	6	5	3	25	75	100
	Core Practical - III (CP)	Database Systems Lab	4	4	3	50	50	100
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	Total		**30**	**22**				**700**

Sem.	Courses	Title	Ins. Hrs	Credits	Exam Hours	Maximum Marks		
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IV	Language Course –IV (LC) - Tamil/Other Languages	Tamil-IV	6	3	3	25	75	100
	English Language Course– IV (ELC)	English-IV	6	3	3	25	75	100
	Core Course–IV(CC)	Big Data Analytics with NoSQL Database	5	5	3	25	75	100
	Core Practical-IV(CP)	Big Data Analytics with NoSQL Database Lab	4	4	3	50	50	100
	Second Allied Course-II(AC)	Mathematics-II	4	3	3	25	75	100
	Second Allied Practical (AP)	Data Analytics Using Spread Sheets Lab	3	2	3	50	50	100
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	Skill Based Elective -II	A. Social Media Analytics B. Business Intelligence	2	2	3	25	75	100
	Extension Activities	NCC,NSS,YRC,YWF	-	1	-	-	-	-
	Gender Studies	Gender Studies	1	1	3	25	75	100
		Total	**30**	**25**				**700**
		Grand Total	**180**	**140**				**4200**

22/5/26
Dr. K. NANDHINI
Asst Professor
Dept of Computer Science
Central University of Tamil Nadu
Thiruvarur

22/05/2026
Dr. K. Umamaheswari
Asst. Professor, Dept. of CS
AGCW, Karaikal
(Affiliated Pondicherry University)

R. Priya/22.5.26
Dr. R. PRIYA
Asst. Professor
Dept. of CSA
Faculty of Science & Humanities
SRMIST
Ramapuram, Chennai

22/05/26
BOS - CHAIRMAN
Department of Artificial Intelligence & Machine Learning, Data Science and Computer Science with Cognitive Systems,
E.G.S.Pillay Arts & Science College (Autonomous),
Nagapattinam - 611 002.

NON MAJOR ELECTIVE–I		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**III**	**A.** Fundamentals of Information Technology **B.** Basics of Office Automation

NON MAJOR ELECTIVE–II		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**IV**	**A.** Working principles of Internet **B.** Web Designing

SKILL BASED ELECTIVE–I		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**V**	**A.** Shell Programming **B.** Digital Marketing

SKILL BASED ELECTIVE–II		
S.No	**SEMESTER**	**TITLE OF THE PAPERS**
1.	**VI**	**A.** Social Media Analytics **B.** Business Intelligence

LIST OF ALLIED COURSES

First Allied Course I — **Statistics**

Second Allied Course II — **Mathematics**

SUMMARY OF CURRICULUM STRUCTURE OF UG PROGRAMMES

Sl. No.	Part	Types of the Courses	No. of Courses	No. of Credits	Marks
1.	I	Language Courses	4	12	400
2.	II	English Courses	4	12	400
3.	III	Core Courses	9	44	900
4.		Core Practical	6	24	600
5.		Allied Courses I & II	4	14	400
6.		Allied Practical	2	4	200
7.		Major Based Elective Courses	2	8	200
8.		Internship	1	2	100
9.		Project	1	3	100
10.	IV	Non-Major Elective Courses	2	4	200
11.		Skill Based Elective Courses	2	4	200
12.		Soft Skills Development	1	2	100
13.		Value Education	1	2	100
14.		Environmental Studies	1	2	100
15.		Health & Wellness	1	1	100
16.	V	Gender Studies	1	1	100
17.		Extension Activities	1	1	--
	Total		**43**	**140**	**4200**

*for those who studied Tamil upto 10th +2 (Regular Stream) + Syllabus for other Languages should be on pass with Tamil at degree level

Those who studied Tamil upto10th +2 but opt for other languages in degree level under Part I should study special Tamil in Part IV

**Extension Activities shall be outside instruction hours

** Internship are 10 to 15 days.

Note:

	Internal Marks	External Marks
1.Theory	25	75
2.Practical	50	50

1. Separate passing minimum is prescribed for Internal and External marks

FOR THEORY

The passing minimum for CIA shall be 40% out of 25 marks [i.e.10 marks]

The passing minimum for University Examinations shall be 40% out of 75 marks [i.e.30 marks]

FOR PRACTICAL

The passing minimum for CIA shall be 40% out of 50 marks [i.e.20 marks]

The passing minimum for University Examinations shall be 40% out of 50 marks [i.e.20 marks]

PROGRAM OUTCOMES (POs)

Upon completion of the B.Sc. Data Science program, graduates will be able to:

PO1: Apply the computing knowledge to analyze real world applications

PO2: Solve the complex problems in the field of data science with an understanding of different types and formats of data.

PO3: Demonstrate a strong foundation in mathematics, statistics, and computational principles for analyzing complex datasets.

PO4: Develop the ability to identify, formulate, and analyze data-driven problems using first principles of mathematics, natural sciences, and domain-specific knowledge.

PO5: Utilize research methodologies to conduct experiments, interpret data, and derive meaningful insights to provide innovative solutions.

Program Specific Outcomes (PSOs)

Graduates of the B.Sc. Data Science program will specifically be able to:

PSO1: Apply machine learning, artificial intelligence, and big data techniques to build predictive and prescriptive models for complex datasets.

PSO2: Implement data science methodologies across domains such as healthcare, finance, marketing, and social sciences.

PSO3: Design, execute, and manage data science projects, demonstrating expertise in handling large datasets, cloud platforms, and advanced programming tools.

PSO4: Proficiently use modern tools and software for data analysis, machine learning, and visualization while understanding their limitations.

PSO5: Commit to lifelong learning to adapt to emerging trends in data science, AI, and related fields, ensuring continual professional growth.

FIRST YEAR	CORECOURSE–I	COURSE CODE: 25DSCC1
SEMESTER- I	PROGRAMMING IN C AND DATA STRUCTURES	CREDIT: 5

COURSE OBJECTIVES:

- To know about the basics of C Programming, Control and Looping Structure and programming with it.
- To understand Arrays, Pointers and String Processing in C language.
- To know about the basic concepts in Data Structures.

UNITS	CONTENTS	HOURS
I	Basic of C: History of C and its importance – Structure of a C program – Data Types – Constants and Variables – Operators and Expressions – Order of Precedence, Evaluating of Arithmetic Expressions – Type Conversion- Decision Statements: if, if-else, and nested if statements.	20
II	Loops Structures: For Loop, While, Do-while loop – Arrays: - One Dimensional Array, Two-dimensional Arrays, Character Arrays and Strings – Functions: Function with arrays- Function with decision and looping statements - Recursion.	20
III	Pointers: Introduction – Pointer Expressions – Chain of Pointers – Pointers and Arrays – Array of Pointers – Pointers as function arguments – Functions returning Pointers – Pointers to Functions – Function pointer – Structures - declaration, initialization, Array of Structures – Pointer to structures, Structures and functions – Typed of Enumerated data types, Unions.	20
IV	File Management in C: Introduction- Defining and Opening a File-Closing a File- Input/output Operations on Files-Error Handling During I/O Operations-Random Access to Files-Command Line Arguments.	15
V	Stack: LIFO concept, Stack operations, Array implementation of stack – Queue: FIFO concept, Queue operations, Array implementation of queue – Singly Linked List: concepts, operations – Doubly Linked List: concepts, operations – Trees: General trees, Binary trees.	15

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	To Summarize the basic knowledge to develop C programs	K2
CO2	To Manipulate Looping, arrays and functions	K4
CO3	To Apply and write programs for solving real world problems	K3, K2
CO4	To Create open, read, manipulate, write and close files.	K5
CO5	To Understand the basic concepts in data structures	K2

TEXTBOOKS:

1. E. Balagurusamy, "Programming in ANSI C", Tata McGraw Hill, New Delhi, Nineth Edition, 2024.
2. E.Horowitz, S.Sahni and Susan Anderson-Freed, "Fundamental Data Structures in C", 2ed, Orient Black Swan Publisher, 2023.
3. Byron S. Gottfried, "Programming with C", Schaum's Outline Series, Tata- McGraw Hill Edition, New Delhi, 2025.

REFERENCES:

1. E. Karthikeyan, "A Textbook on C Fundamentals, Data Structures and Problem Solving", Prentice-Hall of India Private Limited, New Delhi, 2008.
2. Yashavant Kanetkar, "Let us C", BPB Publications, Tenth Edition, New Delhi, 2010.
3. Szuhay, Jeff, and Szuhay, Jeff, "Learn C Programming: A Beginner's Guide to Learning C Programming the Easy and Disciplined Way", Packt Publishing, 2020.
4. Jena, Sisir Kumar, and Jena, Sisir Kumar, "C Programming: Learn to Code", CRC Press, 2021.

WEB RESOURCES:

1. https://www.tutorialspoint.com/cprogramming/index.htm
2. https://www.w3schools.in/data-structures/intro

FIRST YEAR	CORE PRACTICAL-I	COURSE CODE: 25DSCP1
SEMESTER- I	PROGRAMMING IN C LAB	CREDIT: 4

COURSE OBJECTIVES:

- To learn the programming knowledge using C language.
- To create derived and user defined data types.
- To demonstrate the functioning of pointers in C.

LIST OF PROGRAMS

1. Write a Program
    a. To convert temperature from degree Centigrade to Fahrenheit.
    b. To find whether the given number is Even or Odd.
    c. To find the greatest of three numbers.
2. Write a Program to use the switch statement to display Monday to Sunday.
3. Write a Program to display first Ten Natural Numbers and their sum.
4. Write a Program to find Multiplication of Two Matrices.
5. Write a Program
    a. To find the maximum number in Array using pointer.
    b. To reverse a number using pointer.
    c. To add two numbers using pointer.
6. Write a Program to solve Quadratic Equation using functions.
7. Write a Program to find factorial of a number using Recursion.
8. Write a Program to show Call by Value and Call by Reference.
9. Write a Program to create a file containing Student Details.
10. Write a program to implement a stack using singly linked list, Implement Queue using Linked List.

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able:

CO	COURSE OUTCOME	K LEVEL
CO1	To Relate the use of language constructs to solves impel programs	K4
CO2	To Develop programs for various concepts in C language	K4
CO3	To Understand and trace the execution of the list of programs	K2
CO4	To Understand the usage of file handling in C programming Solved at a problems related to data structures	K2
CO5	To Understand the basic concepts in C	K2

E.G.S.PILLAY ARTS & SCIENCE COLLEGE (AUTONOMOUS)
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ALLIED MATHEMATICS

for

B.Sc., Data Science Programme

First Year	ALLIED COURSE - I			Credit:
Semester - I	DESCRIPTIVE STATISTICS			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	6	-	-	6

COURSE OBJECTIVES:

- Understand the significance and computational aspects of statistical analysis.
- Understand the patterns of distribution of data.
- Understand the concept of predictive analytics.

Units	CONTENT	HOURS
Unit I	**Status of statistics and collection of data:** Origin, Scope, Definition, limitations of statistics. Data - Types – Primary and secondary Data – Methods of Data collection.	18
Unit II	**Classification, tabulation and graphical representation:** Classification of data, preparation of tables, diagrammatic representation of data: one dimensional and two dimensional diagram – Graphical representation: Line diagram, Frequency polygon, Frequency curve, Histogram.	18
Unit III	**Measures of central tendency and Location:** Measures of central tendency: Mean, Median, Mode, Geometric mean and Harmonic Mean, Properties with merits and demerits – Empirical relation between means, partition values: Quartiles, Deciles and Percentiles.	18
Unit IV	**Measures of disperssion:** Absolute and relative measures: Range, Mean Deviation, Quartile Deviation and Standard Deviation.	18
Unit V	**Measure of variation, correlation and regression:** Measure of variation: Coefficient of variation – Simple correlation: Karl Pearson's Coefficient of correlation – Rank Correlation- Regression lines.	18

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Understand the scope and necessary of statistics and the representation of data	K1, K2
CO2	Tabulate and represent the data in diagram and Graphs	K2, K3
CO3	Apply the formula and calculate descriptive measure of statistics	K2, K3, K4

CO4	Analyze the nature of data and interpret the measure	K2, K3, K4
CO5	Analyze the data and predict the future values using regression	K4, K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	Agarwal B L (2021) Basic Statistics, New age international Private limited, New delhi, India.
2.	Gupta S.C and Kapoor V.K (2020) Fundamentals of Mathematical Statistics 12th edition, Sultan chand and sons, New Delhi , India.
REFERENCE BOOKS	
1.	Holcomb z. c (2017) Fundamental of Descriptive statistics, Routledge, New york, US
WEB RESOURCES	
1.	https://ntpel.ac.in/coures/111/104/111104120
2.	https://www.iiserpune.ac.in/~bhasbapat/phy221_files/curvefitting.pdf

FIRST YEAR	CORE COURSE - II	COURSE CODE: 25DSCC2
SEMESTER- II	PROGRAMMING IN PYTHON	CREDIT: 5

COURSE OBJECTIVES:

- To develop programs using functions and pass arguments in Python.
- To write programs using loops and decision statements in Python.
- To design and program Python applications.

UNITS	CONTENTS	HOURS
I	Introduction to Python: Features of Python - How to Run Python - Identifiers - Reserved Keywords - Variables - Comments in Python - Indentation in Python - Multi-Line Statements - Multiple Statement Group (Suite) - Quotes in Python - Input, Output and Import Functions - Operators. Data Types and Operations: Numbers – Strings – List – Tuple – Set – Dictionary – Data type conversion.	15
II	Flow Control: Decision Making – Loops – Nested Loops – Types of Loops. Functions: Function Definition – Function Calling - Function Arguments - Recursive Functions - Function with more than one return value.	15
III	Modules and Packages: Built-in Modules - Creating Modules - import Statement - Locating Modules - Namespaces and Scope - The dir() function - The reload() function - Packages in Python - Date and Time Modules. File Handling- Directories in Python.	15
IV	Object-Oriented Programming: Class Definition - Creating Objects - Built- in Attribute Methods - Built-in Class Attributes- Destructors in Python – Encapsulation - Data Hiding – Inheritance - Method Overriding- Polymorphism.	15
V	Exception Handling: Built-in Exceptions-Handling Exceptions- Exception with Arguments - Raising Exception - User-defined Exception- Assertions in Python. Regular Expressions: The match() function - The search() function - Search and Replace - Regular Expression Modifiers: Option Flags-Regular Expression Patterns- Character Classes-Special Character Classes - Repetition Cases - findall() method - compile() method.	15

COURSE OUTCOMES:

Upon successful completion of this course the students would be able:

CO	COURSE OUTCOME	K LEVEL
CO1	To recall and understand the features of python programming language	K2
CO2	To illustrate various programming mechanism used in python	K2
CO3	To apply various language construct to write simple programs in python	K5
CO4	Write java To examine the application of object oriented concept in python programs using Applets	K2
CO5	To distinguish the various constructs used in python.	K5

TEXTBOOKS:

1. Jeeva Jose and P. Sojan Lal, "Introduction to Computing and Problem Solving with PYTHON", Khanna Book Publishing Co, 2016.
2. Mark Summerfield. — Programming in Python 3: A Complete introduction to the Python Language, Addison-Wesley Professional, 2018.
3. Satyanarayana, Radhika Mani, Jagadesh, "Python Programming", Universities Press(India) Pvt Ltd 2024.
4. Wesley J. Chun, "Core Python Programming", Prentice Hall Publication, 2006.

REFERENCES:

1. Timothy A Budd, "Exploring Python", Tata McGraw Hill, New Delhi, 2021
2. Jake Vander Plas, "Python Data Science Handbook: Essential Tools for Working with Data", O'Reilly Media, 2016.
3. Allen B. Downey, ``Think Python: How to Think Like a Computer Scientist, 2nd edition, Updated for Python 3, Shroff/O Reilly Publishers, 2016
4. Guido van Rossum and Fred L. Drake Jr, —An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd., 2021.

WEB RESOURCES:

1. https://www.w3schools.com/python/
2. https://docs.python.org/3/tutorial/index.html

FIRST YEAR	CORE PRACTICAL-II	COURSE CODE: 25DSCP2
SEMESTER - II	PROGRAMMING IN PYTHON LAB	CREDIT: 4

COURSE OBJECTIVES:

- To develop the Numbers, Math functions and Strings.
- To create different Decision Making statements and Functions.
- To design GUI Applications in Python.

LIST OF PROGRAMS

1. Write a python program that displays the following information: Your name, Full address Mobile number, College name, Course subjects.
2. Write a python program to find the largest three integers using if-else and conditional operator.
3. Write a python program that asks the user to enter a series of positive numbers (The user should enter a negative number to signal the end of the series) and the program should display the numbers in order and their sum.
4. Write recursive functions for GCD of two integers.
5. Write recursive functions for the factorial of positive integer.
6. Write recursive functions for Fibonacci sequence up to given number n.
7. Write recursive functions to display prime number from 2 to n
8. Write a python program that writes a series of random numbers to a file from 1 to n and display.
9. Write a python program to sort a given sequence: String, List and Tuple.
10. Write a python program to make a simple calculator.

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able:

CO	COURSE OUTCOME	K LEVEL
CO1	To recall and relate the features of python programming language	K4
CO2	To compare various programming mechanism used in python	K4
CO3	To construct simple programs in python using various language features	K2
CO4	To distinguish the various constructs used in python	K2
CO5	To the application of object oriented concept in python	K2

First Year	ALLIED COURSE - II			Credit: 4
Semester - II	INFERENTIAL STATISTICS			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	5	-	-	5

COURSE OBJECTIVES:

- Gain knowledge on the methods of testing statistical hypothesis.
- Understand the meaning and applications of tests of significance

Units	CONTENT	HOURS
Unit I	**Sampling:** Introduction- Types of sampling- parameter and statistics- sampling distribution.	15
Unit II	**Tests of significance:** Procedure of Testing hypothesis –Standard errors- two types of errors – critical region-Two - tailed and one- tailed tests	15
Unit III	**Tests of significance for large samples:** Sampling of Attributes - test for single proportion - Test of significance for difference of Proportions	15
Unit IV	**Statistical Inference:** Introduction - Characteristics of Estimators – Consistency – Unbiased - Invariance property of consistent Estimators - Sufficient Conditions for Consistency (Simple Problems Only).	15
Unit V	**Sampling Distributions:** Students 't 'distribution – Application of 't' distribution – t- test for single Mean(Simple problems only).	15

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Understand the sampling and distributions	K1, K2
CO2	Understand testing of statistical hypothesis	K2, K3
CO3	Analyze and apply suitable test for analysis.	K2, K3,K4
CO4	Understand the concept of variability to draw inferences	K2, K3,K4
CO5	Understand the concept of powerful tests based on 't' distributions	K4,K5
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	Gupta, S.C., and Kapoor, V. K. (2000). Fundamentals of Mathematical Statistics, 10th revised Edition, Sultan Chand & Sons (Publisher), New Delhi, India.
2.	Gupta S.P (2017), Statistical Methods, 45th revised edition, Sultan Chand & Sons, New Delhi.
REFERENCE BOOKS	

1.	Agarwal, B. L. (2006). Basic Statistics, New Age International Private Limited, New Delhi, India.
WEB RESOURCES	
1.	https://nptel.ac.in/courses/111/104/111104120/

ALLIED PRACTICAL
STATISTICS FOR DATA SCIENCE LAB
(Practical)

COURSE OBJECTIVES:

- To learn practical training in SPSS
- To demonstrate statistics using SPSS
- To analyze various problems using SPSS

1. Find Mean, Median and Mode using SPSS.
2. Calculate Standard deviation and variance using SPSS.
3. Demonstrate Bar diagram in SPSS.
4. Explore t-test for one sample problem.
5. Analyze t-test for two sample problems.
6. Find the Co efficient of correlation using SPSS.
7. Find the Regression equation of X on Y.
8. Formulate the Regression equation of Y on X.
9. Apply One-tailed Test with an example using SPSS.
10. Implement Two-tailed test with a sample problem in SPSS.

COURSE OUTCOMES	
Upon successful completion of this course the students would be able to:	
CO1	Relate the use of SPSS to analyze the problems
CO2	Create charts and plots in SPSS
CO3	Understand the concept of T-test
CO4	Understand the usage of analysis of variance using SPSS
CO5	Analyze problems related to standard deviation

SECOND YEAR	CORE COURSE – III	COURSE CODE: 25DSCC3
SEMESTER - III	DATABASE SYSTEMS	CREDIT: 6

COURSE OBJECTIVES:

- To impart the basic database concepts, applications, data models, schemas and instances.
- To familiarize Entity Relationship model for a database.
- To demonstrate the use of constraints and relational algebra operations.

UNIT	CONTENTS	HOURS
I	Databases and Database Users: Introduction - characteristics of data base approach - intended uses of a DBMS- advantages and implication of database approach. Database Systems Concepts and Architecture: Data models - schemas and instances - DBMS architecture and data independence - database languages and interfaces - database system environment - classification of database management systems.	15
II	Data Modelling :High level conceptual data models for database design- Entity types- entity sets- attributes and keys- Relationships- relationship types- roles and structural constraints - Weak entity types- ER diagrams and design issues.	15
III	Relational Data Model and Relational Algebra: Brief discussion on CODD rules- relational model concepts- constraints and schemas- Update operation on relations- basic and additional relational algebra operations- queries in relational algebra.	15
IV	Structured Query Language (SQL): Data definition in SQL- Basic and complex queries- SQL Insert- delete- update statements and views in SQL- embedded SQL. Database Design: Design guidelines for relational schemas - functional Dependencies - normalization -1st, 2nd, 3rd, 4th and 5th normal forms- Database design process- factors influencing physical database design guidelines and guidelines for relational systems.	15
V	Record Storage and Primary File Organizations: Secondary storage devices - buffering of blocks - placing file records on disk - operations on files - heap files and sorted files – hashing techniques. Index Structure of Files: Single – level and multilevel ordered indexes- dynamic multi-level indexes using B-trees and B+ trees.	15

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand the basic concepts of database systems	K2
CO2	Understand the relational data model	K4
CO3	Apply the concepts of database design.	K3
CO4	Design and create database with normalization	K5
CO5	Analyze the storage and file organization concepts.	K6

TEXTBOOKS:

1. Ramez Elmasri and Shamkant B. Navathe, *"Fundamentals of Database Systems"*, 7th Edition, Pearson Education, 2016.
2. Abraham Silberschatz, Henry F. Korth and S. Sudarshan, *"Database System Concepts"*, 7th Edition, McGraw-Hill Education, 2019.
3. Thomas Connolly and Carolyn Begg, *"Database Systems: A Practical Approach to Design, Implementation and Management"*, 7th Edition, Pearson Education, 2020.

REFERENCES:

1. Hector Garcia-Molina, Jeffrey D. Ullman and Jennifer Widom, *"Database Systems: The Complete Book"*, 2nd Edition, Pearson Education, 2008.
2. David M. Kroenke and David J. Auer, *"Database Processing: Fundamentals, Design and Implementation"*, 16th Edition, Pearson Education, 2023.
3. Alexis Leon and Mathews Leon, *"Database Management Systems"*, Vikas Publishing House, Latest Revised Edition, 2018.

WEB RESOURCES:

1. https://www.w3schools.com/sql/
2. http://infolab.stanford.edu/~ullman/dscb.html
3. https://www.db-book.com/

SECOND YEAR	CORE PRACTICAL – III	COURSE CODE: 25DSCP3
SEMESTER - III	DATABASE SYSTEMS LAB	CREDIT: 4

COURSE OBJECTIVES:

- To understand the basic concepts and the of database systems using MYSQL. applications
- To create and perform basic operation with MYSQL.
- To interact with MYSQL by using nested queries, set of aggregate operations and views.

EXERCISE	CONTENTS
1	Perform the following using SQL: a. Viewing all databases, b. Creating a Database, c. Viewing all Tables in a Database
2	DDL: Creating Tables (With and Without Constraints) a. Primary key, Foreign Key, b. Unique, Default c. Null, Not null d. Enable Constraints, Disable Constraints Drop Constraints
3	DML: Inserting/Updating/Deleting Records in a Table
4	TCL: Saving (Commit) and Undoing (rollback)
5	Perform the following: a. Altering a Table b. Dropping a Table c. Truncating a Table d. Renaming a Table e. Backing up / Restoring a Database
6	For a given set of relation schemes, create tables and perform the following a. Simple Queries, b. Simple Queries with Aggregate functions, c. Queries with Aggregate functions (group by and having clause),
7	For a given set of relation schemes, create tables and perform the following a. Queries involving- Date Functions b. Queries involving -String Functions c. Queries involving - Math Function
8	Perform the following with creating necessary tables. a. Join Queries- Inner Join b. Outer Join c. Sub-queries
9	Perform the following with creating necessary tables a. With IN clause b. With EXISTS clause
10	For a given set of relation tables perform the following a. Creating Views (with and without check option), b. Dropping views, c. Selecting from a view

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Write queries to manipulate data.	K2
CO2	Demonstrate the aggregate functions and set operations.	K4
CO3	Apply the join operations.	K3
CO4	Know about usage of nested subqueries	K5
CO5	Understand the method to create views	K6

Second Year	ALLIED COURSE - III			Credit: 4
Semester - III	ALLIED MATHEMATICS I			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	6	-	-	6

COURSE OBJECTIVES:

- To revise the basic concepts of probability, axiomatic theory of probability.
- To understand the concept of random variable.
- Understand the graph, and graph models, terminology of graph.

Units	CONTENT	HOURS
Unit I	**Theory of Probability:** Introduction – Definition of various terms – Mathematical/ Classical Probability – Statistical/ Empirical Probability – Operations on Sets – Algebra of Sets – Probability Functions.	18
Unit II	**Multiplication law of probability and Conditional Probability:** Definition of conditional probability of an event - Results on conditional probability - Definition of independence of two events $(A \cap B) = (A)P(B)$ - Pairwise independence and mutual independence for three events - Multiplication theorem $P(A \cap B) = P(A)\ P(B\|A)$ - Bayes' theorem.	18
Unit III	**Random Variables and Distribution functions:** Random Variables - Distribution functions – Properties of Distribution function – Discrete random variable – Probability mass function- Discrete Distribution function (Simple problems only).	18
Unit IV	**Graphs and subgraphs:** Introduction - Definition and Examples - Degrees - Subgraphs - Isomorphism. –independent sets and coverings.	18
Unit V	Matrices - Operations on Graphs - Walks, Trails and Paths – Connectedness and Components - Eulerian Graphs.	18

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Explain the basic concepts of probability and sets.	K2
CO2	Apply conditional probability and Bayes' theorem to solve problems	K3
CO3	Solve simple problems on random variables and distribution functions	K3
CO4	Analyze graphs using degrees, subgraphs, and isomorphism	K4
CO5	Apply graph operations to study walks, paths, connectedness, and Apply graph operations to study walks, paths, connectedness, and Eulerian graphs.	K3
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	Gupta, S. C. and Kapoor, V. K. (1983). Fundamentals of Mathematical Statistics, Eighth Edition, Sultan Chand and Sons Publishers, New Delhi.
2.	S. Arumugam and S. Ramachandran, Invitation to Graph Theory, SciTech Publications (India) Pvt. Ltd., Chennai, 2006.

REFERENCE BOOKS	
1.	Rao, VLS Prakash (2008). First Course in Probability and Statistics, New Age International Publishers, New Delhi.
WEB RESOURCES	
1.	https://nptel.ac.in/

SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25DSNM1A
SEMESTER - III	FUNDAMENTALS OF INFORMATION TECHNOLOGY	CREDIT: 2

COURSE OBJECTIVES:

- To familiarize the students with the world of IT and IT-enabled services.
- To provide an in-depth knowledge about internet and internet tools.
- To enable the students to understand about Computer Security.

UNIT	CONTENTS	HOURS
I	Introduction to Computers - Generation of Computers - Classification of Digital Computer - Anatomy of Digital Computer.	5
II	CPU and Memory - Secondary Story Devices - Input Devices - Output Devices.	5
III	Introduction to Computer Software - Programming Language - Operating Systems - Introduction to Database Management System.	5
IV	Computer Networks - WWW and Internet - Email - Web Design	5
V	Computers at Home, Education, Entertainment, Science, Medicine and Engineering - Introduction to Computer Security - Computer Viruses, Bombs, Worms.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand basic concepts and terminologies in IT and IT-enabled services.	K2
CO2	Understanding personal computers and their operations.	K2
CO3	Understand about operating systems and database management	K2
CO4	Comprehend about WWW, internet, email and web design concepts	K4
CO5	Respond to computer security issues.	K3

REFERENCES:

1. Alexis Leon and Mathews Leon, Fundamentals of Information Technology, Vikas Publishing House Pvt. Ltd, 2009
2. Fundamentals of Computers and Information Technology, M.N Doja,2005
3. Ramesh Bangia,"Computer Fundamentals and Information Technology", Laxmi Publications Pvt Limited, 2008.
4. Bharihoke, "Fundamentals of Information Technology", Excel Books, 2009. Ralph Stair, George Reynolds,
5. "Fundamentals of Information Systems" Cengage Learning, 2015.

SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25DSNM1B
SEMESTER - III	BASICS OF AUTOMATION	CREDIT: 2

COURSE OBJECTIVES:

- To familiarize students with database concepts and data handling using MS-Access.
- To develop practical skills in using MS-Office applications for documentation & database management.
- To provide knowledge of spreadsheet operations, formulas, charts, and data analysis using MS-Excel.

UNIT	CONTENTS	HOURS
I	Introduction to MS – Windows, Operating System, Windows Desktop Icons and their functions, Dialog Boxes, Task Bar, Parts of Windows, Keyboard short keys or hot keys. Introduction to MS-Office.	5
II	MS-Word – creating, editing saving and printing text documents, Font formatting, inserting tables, smart arts, page breaks, working with images, Spelling and Grammar Check, document properties, Mail Merge.	5
III	MS-Excel – spreadsheet basics, creating, editing saving and printing spreadsheets, working with functions & formulas, modifying worksheets with color & auto formats, graphically representation : charts & graphs, data forms, analyzing data – securing spreadsheets	5
IV	MS-Power Point – creating, editing saving and printing slides, auto layouts, custom animation, slide transition, Graphical representation – charts & graphs, slide presentation.	5
V	MS-Access – planning a Database, access screen, creating database, tables, working with forms, creating queries, types of reports, printing & print preview, importing data.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand the basic concepts of operating systems, Windows environment, and MS-Office tools.	K2
CO2	Create, edit, format, and print professional documents using MS-Word.	K3
CO3	Prepare spreadsheets, apply formulas and functions, analyze data, and generate charts using MS-Excel.	K3
CO4	Create and manage simple databases, forms, queries, and reports using MS-Access.	K3
CO5	Apply office automation tools effectively for academic, personal, and professional purposes.	K3

REFERENCES:

1. Computer Fundamental & MS Office — Shashwat Publication, 2025 Edition. Covers Windows, MS Word, Excel, PowerPoint and Access aligned with NEP-2020 syllabus.
2. Technology For Success & The Shelly Cashman Series Microsoft 365 & Office Introductory — Cengage, 2025 Edition. Updated for Microsoft 365 and Office 2021 applications.
3. Keyboard 3: Windows 10 and MS Office 2016 — Oxford University Press India, 2025 Edition. Useful for Office automation fundamentals and practical exercises

	COURSETITLE	YEAR	SEMESTER	CREDIT
	A.FOOD AND NUTRITIONAL BIOTECHNOLOGY	II	III	2

OBJECTIVES

- The students will be able to learn about food Preservation and microorganisms associated with itLearn about packaging of different foods. Learn types of Foods and their nutritional values
- Learn about components of foods and their deficiency disorders.5. Learn about Essential minerals , BMR and RDA

Unit I

Introduction to Food and Nutrition Biotechnology, Application of biotechnology to improve food quality, safety, production, and Nutritional value. Nutritional needs of the growing population.

Unit II

Food Fermentation Technology- Use of microorganisms to produce and preserve food products. Fermented foods - curd, cheese, bread, and vinegar have improved taste, shelf life, and nutritional value.

Unit III

Food Processing and Preservation- Food processing and preservation techniques help maintain food quality and prevent spoilage. Biotechnology contributes to the development of safer and longer-lasting food products through various preservation methods.

Unit IV

Nutritional Improvement of Foods Biotechnology is used to enhance the nutritional content of food through biofortification and genetic modification. Nutrient-enriched crops help reduce deficiencies and improve public health

Unit V

Functional foods and nutraceuticals - provide health benefits beyond basic nutrition, bioactive compounds that help prevent diseases and promote overall health and Well being.

REFERENCES

1. "Food Biotechnology "by Elsayed Abdel-Aaland And yKhatwa (2019)

2. "Introduction to Food Biotechnology" by Perry Johnson-Green(2016)

3. "Food Biotechnology "by Kalidas Shetty and Gopinadhan Paliyath (2005)

4. "Food Biotechnology, Second Edition "by Klaus Buchholz and Volker Kasche (2013)

5. "Nutritional Biochemistry and Metabolism: With Clinical Applications"by Maria Luz Fernandez and Jose M. Ordovas (2014)

6. "Biotechnology in Functional Foods and Nutraceuticals" edited by Debasis Bagchi and Francis, C. Lau (2010)

OUTCOMES

Understand food and nutrition biotechnology principles.

Explain microbial and enzymatic applications in food processing.

Apply biotechnology for food quality improvement.

Analyze nutritional and health benefits of foods.

Evaluate functional foods and nutraceuticals.

Understand food safety and modern biotechnological developments.

SECOND YEAR	CORE PRACTICAL – IV	COURSE CODE: 25DSCP4
SEMESTER - IV	BIG DATA ANALYTICS WITH NoSQL DATABASE LAB	CREDIT: 4

COURSE OBJECTIVES:

- To explain big data concepts.
- To identify the Hadoop eco systems.
- To experience on various NoSQL Databases

EXERCISE	CONTENTS
1	Set up a pseudo-distributed, single-node Hadoop cluster backed by the Hadoop Distributed File System
2	Installation of Hadoop Ecosystem - Pig
3	Installation of Hadoop Ecosystem - Hive
4	Write a hive query to load the data into table from any local file on the system.
5	Write a hive query, which returns the contents of the whole table.
6	Demonstrate the usage of Hive functions.
7	Installation of MongoDB
8	Reading CSV file and loading into MongoDB
9	Reading JSON file and Loading into MongoDB
10	Reading MongoDB and writing into MySQL

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Install Hadoop distributed file system	K5
CO2	Install Hadoop ecosystems	K5
CO3	Write Hive queries	K2
CO4	Implement MongoDB for unstructured data	K4
CO5	Use read and write various files in MongoDB	K5

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SECOND YEAR	CORE PRACTICAL – IV	COURSE CODE: 25DSCP4
SEMESTER - IV	BIG DATA ANALYTICS WITH NoSQL DATABASE LAB	CREDIT: 4

COURSE OBJECTIVES:

- To explain big data concepts.
- To identify the Hadoop eco systems.
- To experience on various NoSQL Databases

EXERCISE	CONTENTS
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2	Installation of Hadoop Ecosystem - Pig
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4	Write a hive query to load the data into table from any local file on the system.
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6	Demonstrate the usage of Hive functions.
7	Installation of MongoDB
8	Reading CSV file and loading into MongoDB
9	Reading JSON file and Loading into MongoDB
10	Reading MongoDB and writing into MySQL

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Install Hadoop distributed file system	K5
CO2	Install Hadoop ecosystems	K5
CO3	Write hive queries	K2
CO4	Implement MongoDB for unstructured data	K4
CO5	Use read and write various files in MongoDB	K5

Second Year	ALLIED COURSE - IV			Credit: 4
Semester - III	ALLIED MATHEMATICS II			Subject Code:
Instructional Hours per week	Lecture	Tutorial	Practical	Total
	6	-	-	6

COURSE OBJECTIVES:

- To understand the basic arithmetic operations on vectors and matrices, including determinants, using technology where appropriate.
- To understand find the eigen values and eigenvectors of a matrix or a linear transformation, and using them to diagonalize a matrix.
- To introduce fundamental concepts and terminology of discrete mathematics.

Units	CONTENT	HOURS
Unit I	**System of Linear Equation:** Matrices, Determinants, Cramer's Rule, Echelon form, Row reduction, Gaussian elimination Method.	18
Unit II	**Vector Spaces:** Introduction to vector spaces, Some properties of vector spaces, Linear combination, Linear independence, Linear dependence, Basis and Dimension of a vector space, Row space, Column space.	18
Unit III	**Eigen values and Eigen vectors:** Eigen values and Eigen vectors, The characteristic equation, Diagonalization.	18
Unit IV	**Relations**: Introduction - Types of Relations, Equivalence relations, Digraphs of relations, Matrix representation and Composition of Relations, Poset, Hasse diagram.	18
Unit V	**Boolean Functions :** Introduction, Boolean variable, Boolean Function of degree n, Boolean identities, Definition of Boolean Algebra, Representation of Boolean Functions : Minterm, Maxterm.	18

COURSE OUTCOMES		COGNITIVE LEVEL
CO1	Solve systems of linear equations using methods by Gaussian elimination.	K3
CO2	Demonstrate understanding of the concepts of vector space, linear independence and basis.	K2
CO3	Determine eigen values and eigenvectors and solve eigenvalue problems.	K3
CO4	Demonstrate understanding the use of truth tables and laws of identity, distributive, commutative, and domination.	K3
CO5	Simplify and prove Boolean expressions, Compute sum of products and product of sum expansions.	K4
K1 - Remember; **K2** - Understand; **K3** - Apply; **K4** - Analyze; **K5** - Evaluate; **K6** – Create.		

TEXTBOOKS	
1.	K. Hoffman and R. Kunze, Linear Algebra, 2nd edition(2014), Prentice Hall of India, New Delhi.
2.	Dr Swapan kumar sarkar, A textbook of Discrete Mathematics, S. Chand Publishing, 2016.
REFERENCE BOOKS	

1.	Howard Anton, Chris Rorres, Elementary Linear Algebra, Application Version, Ninth Edition, Wiley, 11th edition.
2.	T. Veerarjan, Discrete Mathematics, McGraw Hill Companies, 2012.
WEB RESOURCES	
1.	https://nptel.ac.in/

ALLIED PRACTICAL

ALLIED MATHEMATICS (MAXIMA SOFTWARE)

(Practical)

COURSE OBJECTIVES:

- To solve systems of linear equations, using software to facilitate row reduction.
- To understand find the eigenvalues and eigenvectors of a matrix and using them to diagonalize a matrix.
- To know how to use maxima software.

EXPERIMENTS:

1. Cramer's Rule of solving system of linear equations
2. Gauss Elimination method of solving system of linear equations
3. Find the basis and dimension of a vector space
4. Check linear independence using determinants.
5. Find the determinant and inverse of a matrix
6. Find the eigen values and eigen vectors of a matrix.
7. Check whether a relation is a equivalence relation.
8. Find the matrix representation of a relation.
9. Generate truth table of a Boolean function.
10. Display minterms and maxterms for three variables.

COURSE OUTCOMES	
Upon successful completion of this course the students would be able to:	
CO1	Understand the systems of linear equations using methods by Gaussian elimination.
CO2	Demonstrate understanding of the concepts of vector space, linear independence and basis.
CO3	Compute eigenvalues and eigenvectors problems.
CO4	Demonstrate the use of truth tables and laws of identity, distributive, commutative, and domination.
CO5	Simplify and prove Boolean expressions, Compute sum of products and product of sum expansions.
CO6	Students can solve the problem based on theory by using maxima software.

SECOND YEAR	NON-MAJOR ELECTIVE- II	COURSE CODE: 25DSNM2A
SEMESTER - IV	WORKING PRINCIPLES OF INTERNET	CREDIT: 2

COURSE OBJECTIVES:

- To teach the basics of the World Wide Web
- To understand the fundamentals of the Internet and the usage
- To know the components of Multimedia on the internet

UNITS	CONTENTS	HOURS
I	What is Internet? The Internet's underlying Architecture	5
II	Connecting to the Internet – Communicating on the Internet	5
III	How the World Wide Web works. Common Internet tools	5
IV	Multimedia on the Internet – Intranet and shopping on the Internet	5
V	Safeguarding the Internet	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Understand the evolution of the Internet.	K2
CO2	Know the basic knowledge of the web	K4
CO3	Comprehend the protocols and standards used throughout the Internet.	K2
CO4	Discuss a variety of Internet and WWW applications and related technologies.	K4
CO5	Evaluate the opportunities and threats created by interconnecting computers via the Internet.	K5

REFERENCES:

1. Preston Gralla, "How the Internet Works", Pearson Education, Eighth Edition, 2006.
2. C.Xavier, Fundamentals of Internet and Emerging Technologies, New Age International Private Limited; First Edition ,2021
3. Alexis Leon, Internet for Everyone, S. Chand (G/L) & Company Ltd; Second Edition 2012.
4. Andrea C. Nakaya,"Internet and Social Media Addiction", Reference Point Press,2015.
5. Richard Fox, Wei Hao,"Internet Infrastructure: Networking, Web Services, and Cloud Computing", CRC Press, 2017.

SECOND YEAR	NON MAJOR ELECTIVE II	COURSE CODE: 25DSNMS2B
SEMESTER - IV	WEB DESIGNING	CREDIT: 2

COURSE OBJECTIVES:

- To understand the basic concepts of computer networks, client–server architecture, and web technologies.
- To gain knowledge of web browsing, search engines, email systems, and web navigation techniques.
- To design web pages using HTML including text formatting, lists, images, links, and tables.

UNIT	CONTENTS	HOURS
I	Introduction to Web related terms: Client –Server–Single Tier Architecture–Two Tier Architecture– N Tier Architecture – Internet – Intranet – WWW – Web Page – Web Site.	5
II	Introduction to Internet and its Protocols: Internet Addressing – IP Address – Domain Name – Electronic Mail–Uniform Resource Locator	5
III	Web Browser: Web Index – Web Search Engine–Meta Search Sites–Directory and Indexes – Managing Mails: Address Book – File Attachment Facility – Junk Mail	5
IV	Introduction to HTML: HTML Code for Web Page–HTML Tags-Web Page Basics–Display a Web Page in a Web Browser – Format texts – Font Style and sizes – Lists – Ordered Lists – Unordered Lists – Image – Add an image – Background Image – Horizontal Rule	5
V	Introduction to HTML Table: Cell Padding – Cell Spacing – Table Borders, HTML–Advantages and Disadvantages.	4

COURSE OUTCOMES:

Upon successful completion of this course, the students would be able to:

CO	COURSE OUTCOME	K LEVEL
CO1	Explain the fundamentals of Internet, WWW, web architecture, and networking concepts.	K1
CO2	Identify and describe various Internet protocols and their roles in web communication.	K3
CO3	Use web browsers, search engines, and email systems effectively for information retrieval and communication.	K3
CO4	Design basic web pages using HTML elements such as text formatting, lists, images, and links.	K4
CO5	Create and manage HTML tables and apply formatting techniques for structured data representation.	K5

REFERENCES:

1. Alexis Leon & Mathews Leon, "Internet for Everyone", Leon Tech World, Chennai.2, 2009
2. John Zabour,JeffFoust & David Kerven,"HTML 4 HOW-TO",2005
3. Eric Kramer,"HTML". Kamalesh N.Agarwala, AmitLal & Deeksha Agarwala,"Business of the net, 2008.
4. Responsive Web Design with HTML5 and CSS (5th Edition) – *Ben Frain, Packt Publishing, 2025*

COURSE CODE	COURSE TITLE	YEAR	SEMESTER	CREDIT
	BIOTECHNOLOGY FOR HUMAN HEALTH	II	IV	2

Objectives:

- To develop new medicines, vaccines, and diagnostic methods for diseases
- To improve human health through genetic engineering and modern biotechnology techniques
- To prevent and treat infectious and genetic diseases effectively

Unit I

Industry Enzymes for textile industry, breweries, food supplements – single cell protein, vitamins, food processing - cheese, yoghurt making, biodegradable plastics, biofuels.

Unit II

Environment Applications of Biotechnology in environmental aspects: waste management, biodegradation of heavy metals, water cleaning, removing oil spills, air and soil pollution, bioremediation, biomining.

Unit III

Environment management - Environmental Impact Assessment - waste minimization techniques - environmental planning in urban development - natural resources and sustainable development - environmental ethics. tion, Increased milk production, artificial insemination, poultry and fisheries.

Unit IV

Applications in livestock improvement: transgenic animals, animal vaccine production .Transgenic crops for human

Unit V

Human Health and livestock Applications .Antibiotic production, Molecular diagnostics, vaccines and vaccine delivery, recombinant therapeutics – insulin, gene therapy, forensics.

Text Books/References

1. Bhasin, M.K. and Nath, S. 2002. Role of Forensic Science in the New Millennium, University of Delhi,

2. Crueger Wand Crueger, A. 2000. Biotechnology: A textbook of Industrial Microbiology. 2nd edition. Panima Publishing Co. New Delhi.

3. Eckert, W.G. and Wrightin, R.K. 1997. Introduction to Forensic Sciences, 2nd Edition, CRC Press, Boca Raton.

4. Hans-Joachim Jordening and Jesef Winter, 2005. Environmental Biotechnology Concepts and Applications.

5. James, S.H. and Nordby, J.J. 2005. Forensic Science: An Introduction to Scientific and Investigative Techniques, 2nd Edition, CRC Press, Boca Raton.